SI Session Planning Form

SI Leader: Brandon Cobb
Session Week: 6
Session#: 6
Instructor: Professor Tramontozzi
Old Plans: SharePoint
Session Goal: Students will have reviewed unit conversions, algebraic manipulation of exact and inexact numbers, calculating their significant figures after multi-step operations like addition and multiplication. Student will review how to write scienticific notation. Students are given a reference sheet for conversion factors. Students will become adept with atomic particles and subatomic structures like orbitals of electrons.

Opener

Take Attendance

Collaborative Learning Techniques	Group Discussion
Learning Strategy	Note Review
Time	12:00—12:10
Materials	Notes, homework, whiteboard, markers
Lecture/Textbook/Old Plans/Slide References	Class roster, Chapter 3: Periodic Table, Chapter 4: Chemical Bonding: The Ionic Bond Model, Chapter 5: The Chemical Bonding: The Covalent Bond Model

Definitions

1. **Group Discussion:** One person in the group is asked to present on a topic or review material for the group and then lead the discussion for the group. This person should not be the regular group leader.
2. **Summarize Lecture:** As a group summarize the lecture from the previous class. You may have to provide prompts for the students. For example, "The first concept discussed was Civil Liberties adn Public Policy, what did the profressor highlight regarding this?" You may want to ask them to try summarizing without looking at their notes; however, if they are having a difficult time remembering, tell them to refer to their notes.

Instructions (please provide aDETAILED activity breakdown below (and/or attached

Group Discussion (Test Revuew):

1. The SI leader forms small groups of 3–5 students.
2. Each group is asked to take 2–3 minutes to quickly scan their notes, homework, or past practice and identify at least two questions or topics they want clarified. These can be areas of confusion, tricky rules, or "why does this work this way?" type questions.
3. Each group writes their questions on a shared whiteboard, large paper, or digital board so the whole class can see.
4. The SI leader facilitates by clustering similar questions together and prompting groups to tackle them collaboratively.
5. Groups take turns explaining their reasoning for the questions they selected. Other groups may add clarifications, alternative explanations, or corrections.
6. If disagreements or misconceptions arise, the leader steps in to guide discussion, referencing key concepts rather than just giving direct answers.
7. The session continues until most major question categories have been addressed. The leader highlights connections between topics (e.g., atomic structure → electron configuration → bonding → polarity → naming).
8. At the end, students individually write down one "aha!" insight they gained from peers and one question they still want to follow up on to help focus future studying.

Checking for Understanding (questions & answers) continued...

1. What makes protons, neutrons, and electrons different in role and importance?
2. Why does changing the number of protons change the identity of the atom?
3. Why do electrons contribute so little to atomic mass?
4. What does the "mass number" actually represent conceptually?
5. Why can the same element exist with different mass numbers?
6. Why do isotopes have nearly identical chemical behavior even if their masses differ?
7. Why is the average atomic mass often not a whole number?
8. Why are elements arranged in periods and groups instead of just one long list?
9. What does being in the same group tell you about an element's behavior?
10. Why does reactivity often change in predictable patterns across a period?
11. Why do electrons fill orbitals in a particular order instead of randomly?
12. What's the purpose of Hund's rule when filling orbitals?
13. Why do some electron configurations have exceptions to the predicted order?
14. How does electron configuration connect to the position of an element on the periodic table?
15. Why do atoms gain or lose electrons to form ions?
16. Why are certain charges "fixed" for some metals but variable for others?
17. Why do nonmetals typically form negative ions while metals form positive ones?
18. Why do atoms form bonds in the first place?
19. Why do some atoms share electrons equally while others don't?
20. Why do some atoms or ions have the same electron configuration as noble gases?
21. Why do some ions consist of multiple atoms bonded together instead of just one charged atom?
22. Why does the number of electron groups determine molecular shape?
23. Why do lone pairs of electrons affect molecular shape differently than bonding pairs?
24. Why do molecules with the same number of electron groups sometimes have different shapes?
25. Why do atoms differ in their ability to attract electrons?
26. Why does a bond become polar if atoms pull unequally on shared electrons?
27. Why can a molecule with polar bonds still be overall nonpolar?
28. Why is molecular shape important in determining polarity?

Activity 1

Collaborative Learning Techniques	Jigsaw
Learning Strategy	Predict Test Questions
Time	12:10—12:30
Materials	Notes
Lecture/Textbook/Old Plans/Slide References	Chapter 3: Periodic Table, Chapter 4: Chemical Bonding: The Ionic Bond Model, Chapter 5: The Chemical Bonding: The Covalent Bond Model

Definition

1. **Jigsaw:** Jigsaws, when used properly, make the group as a whole dependent upon all of them in subgroups. Each group provides a peice of the puzzle. Group members are broken into smaller groups. Each small group works on some aspect of the same problem, question, or issue. They then share their part of the puzzle with the large group.
2. **Predict Test Questions:** Put students in groups of 2 or 3 and assign them to write a test question for a specific topic, ensuring that all topics have been convered. Ask students to write their questions on the board or on an overhead for discussion (would the professor ask the question? What is the answer? etc.). Students will have the benefit of learning to think like the teacher and they'll be able to see additional questions that other students have writen.

Instructions (please provide aDETAILED activity breakdown below (and/or attached

1. **Distribute Worksheets:** Each student receives a worksheet containing chemistry questions. The problems cover a range of difficulty so that students can decide which ones best match their needs.
2. **Student Choice of Starting Point:** Students look over the worksheet and select their own starting problem rather than being assigned a fixed order. This allows them to challenge themselves where they feel it is most beneficial.
3. **Staggered Timing:** Students work in pairs. Each partner begins at a different point (for example, one may start at Problem 1 while the other starts at Problem 3). They continue by alternating every three questions. This staggered structure ensures that students will be asking for help or requesting solutions at different times, keeping the class flow balanced.
4. **Independent Work:** Students attempt their chosen problems individually, writing down both reasoning and answers. Emphasis is placed on documenting thought process as well as the result.
5. **Help and Solution Reveal:** After the set time for each problem expires, the SI leader either walks through the solution on the whiteboard or directs students to check the answer key. Students are encouraged to ask clarifying questions at the timed checkpoints rather than all at once.
6. **Partner Collaboration:** Once a solution is reviewed, partners share their reasoning for their respective problems. They ask each other questions, offer corrections, and compare how their approaches aligned with the solution.
7. **Rotation of Problems:** After every three questions, students switch to new problems, continuing the staggered order so that they stay offset from each other. This ensures that each student engages with a variety of problems across the worksheet.
8. **Class-Wide Debrief:** The SI leader pauses periodically to highlight frequently asked questions, difficult concepts, or common errors observed during the staggered work. Whole-class solutions are discussed when many students identify the same challenge.
9. **Reflection:** At the close, students note which problems gave them the most insight, what strategies helped, and what topics they still want clarified. This reflection helps focus future review sessions.

Content (fully worked out problems & answers) continued...

(1) Calculate the atomic mass of carbon given its common isotopes: ^{12}C has mass 12.000000 amu (98.930%) and ^{13}C has mass 13.003355 amu (1.070%).

a. 12.001 amu

b. 12.01 amu

c. 12.10 amu

d. 13.003 amu

Solution:
Convert abundances to decimals: 0.9893 and 0.0107.

$$\text{atomic mass} = (12.000000)(0.9893) + (13.003355)(0.0107)$$

Compute each term:

$$(12.000000)(0.9893) = 11.87$$

$$(13.003355)(0.0107) = 0.1391$$

$$\text{atomic mass} = 11.87 + 0.1391 = 12.01 \text{ amu}$$

Answer: b. 12.01 amu

Content (fully worked out problems & answers) continued...

(2) Calculate the atomic mass of chlorine given its common isotopes: ^{35}Cl mass = 34.96885 amu (75.78%), ^{37}Cl mass = 36.96590 amu (24.22%).

a. 35.00 amu

b. 35.178 amu

c. 35.45 amu

d. 36.000 amu

Solution:
Convert abundances: 0.7578 and 0.2422.

$$\text{atomic mass} = (34.96885)(0.7578) + (36.96590)(0.2422)$$

Compute each term:

$$(34.96885)(0.7578) = 26.50$$

$$(36.96590)(0.2422) = 8.950$$

$$\text{atomic mass} = 26.50 + 8.950 = 35.45 \text{ amu}$$

Answer: c. 35.45 amu

Content (fully worked out problems & answers) continued...

(3) Calculate the atomic mass of magnesium using three common isotopes: ^{24}Mg mass = 23.985042 amu (78.99%), ^{25}Mg mass = 24.985837 amu (10.00%), ^{26}Mg mass = 25.982593 amu (11.01%).

a. 24.31 amu

b. 24.000 amu

c. 25.000 amu

d. 23.985 amu

Solution:
Convert abundances: 0.7899, 0.1000, 0.1101.

$$\text{atomic mass} = (23.985042)(0.7899) + (24.985837)(0.1000) + (25.982593)(0.1101)$$

Compute each term:

$$(23.985042)(0.7899) = 18.95$$

$$(24.985837)(0.1000) = 2.499$$

$$(25.982593)(0.1101) = 2.861$$

$$\text{atomic mass} = 18.95 + 2.499 + 2.861 = 24.31 \text{ amu}$$

Answer: a. 24.31 amu

Content (fully worked out problems & answers) continued...

(4) Calculate the atomic mass of silver from its two stable isotopes: ^{107}Ag mass = 106.90509 amu (51.839%), ^{109}Ag mass = 108.90476 amu (48.161%).

a. 107.000 amu

b. 107.868 amu

c. 108.000 amu

d. 106.905 amu

Solution:
Convert abundances: 0.51839 and 0.48161.

$$\text{atomic mass} = (106.90509)(0.51839) + (108.90476)(0.48161)$$

Compute each term:

$$(106.90509)(0.51839) = 55.427$$

$$(108.90476)(0.48161) = 52.441$$

$$\text{atomic mass} = 55.427 + 52.441 = 107.868 \text{ amu}$$

Answer: b. 107.868 amu

Content (fully worked out problems & answers) continued...

(5) Calculate the atomic mass of bromine given: ^{79}Br mass = 78.918337 amu (50.690%), ^{81}Br mass = 80.916291 amu (49.310%).

a. 79.000 amu

b. 80.500 amu

c. 79.903 amu

d. 78.918 amu

Solution:
Convert abundances: 0.50690 and 0.49310.

$$\text{atomic mass} = (78.918337)(0.50690) + (80.916291)(0.49310)$$

Compute each term:

$$(78.918337)(0.50690) = 39.999$$

$$(80.916291)(0.49310) = 39.904$$

$$\text{atomic mass} = 39.999 + 39.904 = 79.903 \text{ amu}$$

Answer: c. 79.903 amu

Content (fully worked out problems & answers) continued...

(6) Which group of the periodic table do the alkali metals belong to?

a. Group 1

b. Group 2

c. Group 17

d. Group 18

Solution:
Alkali metals include Li, Na, K, Rb, Cs, and Fr. These elements are found in the first vertical column of the periodic table, which is Group 1.

Answer: a. Group 1

Content (fully worked out problems & answers) continued...

(7) Calcium (Ca) is classified as which type of element?

a. Alkali metal

b. Alkaline earth metal

c. Halogen

d. Noble gas

Solution:
Calcium (atomic number 20) is found in Group 2 of the periodic table. Group 2 elements are called alkaline earth metals.

Answer: b. Alkaline earth metal

Content (fully worked out problems & answers) continued...

(8) Which period does chlorine (Cl) belong to?

a. Period 2

b. Period 3

c. Period 4

d. Period 5

Solution:
Chlorine (atomic number 17) has its outer electrons in the 3rd energy level, so it belongs to Period 3.

Answer: b. Period 3

(9) Which of the following is a noble gas?

a. Oxygen (O)

b. Fluorine (F)

c. Neon (Ne)

d. Sodium (Na)

Solution:
Noble gases are found in Group 18. Neon (Ne) is in Group 18, while oxygen and fluorine are nonmetals in Groups 16 and 17, and sodium is an alkali metal in Group 1.

Answer: c. Neon (Ne)

Content (fully worked out problems & answers) continued...

(10) Which element is a metalloid?

a. Boron (B)

b. Magnesium (Mg)

c. Argon (Ar)

d. Bromine (Br)

Solution:
Metalloids have properties of both metals and nonmetals. Boron (B) lies along the staircase line and is classified as a metalloid.

Answer: a. Boron (B)

Content (fully worked out problems & answers) continued...

(11) Which group contains the halogens?

a. Group 1

b. Group 2

c. Group 17

d. Group 18

Solution:
The halogens include fluorine, chlorine, bromine, iodine, and astatine. These are located in Group 17 of the periodic table.

Answer: c. Group 17

Content (fully worked out problems & answers) continued...

(12) Which of the following is a transition metal?

a. Iron (Fe)

b. Sodium (Na)

c. Sulfur (S)

d. Neon (Ne)

Solution:
Transition metals are found in Groups 3–12 of the periodic table. Iron (Fe) is in Group 8, making it a transition metal.

Answer: a. Iron (Fe)

Content (fully worked out problems & answers) continued...

(13) Which element is in Group 2, Period 4?

a. Calcium (Ca)

b. Magnesium (Mg)

c. Beryllium (Be)

d. Strontium (Sr)

Solution:
Group 2 elements are alkaline earth metals. In Period 4, the Group 2 element is Calcium (Ca).

Answer: a. Calcium (Ca)

Content (fully worked out problems & answers) continued...

(14) Which of the following elements is a nonmetal?

a. Aluminum (Al)

b. Phosphorus (P)

c. Iron (Fe)

d. Calcium (Ca)

Solution:
Phosphorus (P) is a nonmetal located in Group 15. The others are metals: aluminum (metal), iron (transition metal), and calcium (alkaline earth metal).

Answer: b. Phosphorus (P)

Content (fully worked out problems & answers) continued...

(15) Which element is in Period 2, Group 18?

a. Helium (He)

b. Neon (Ne)

c. Argon (Ar)

d. Krypton (Kr)

Solution:
Period 2 corresponds to the second row of the periodic table. In Group 18 of this row, the element is Neon (Ne).

Answer: b. Neon (Ne)

Content (fully worked out problems & answers) continued...

(16) Write both the longhand and shorthand (noble-gas) electron configurations for neutral chromium (Cr).

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$; Shorthand: [Ar]$4s^2 3d^4$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$; Shorthand: [Ar]$4s^1 3d^5$

c. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^6$; Shorthand: [Ar]$3d^6$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$; Shorthand: [Ar]$4s^2 3d^3$

Solution:
Chromium, Cr, has atomic number $Z = 24$ so there are 24 electrons in the neutral atom. Fill orbitals in order (Aufbau) but note the well-known exception for Cr: a half-filled d subshell+a single s electron is more stable than $4s^2 3d^4$. Thus electrons arrange as:
Longhand (write from lowest to highest energy):

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$$

Check electron count: $2 + 2 + 6 + 2 + 6 + 1 + 5 = 24$.
Shorthand: the core through 3p is argon: [Ar]. So shorthand is: [Ar]$4s^1 3d^5$.

Answer: b. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$; [Ar]$4s^1 3d^5$

Content (fully worked out problems & answers) continued...

(17) Write both the longhand and shorthand electron configurations for Fe^{3+}.

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$; Shorthand: [Ar]$3d^6$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$; Shorthand: [Ar]$4s^1 3d^5$

c. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$; Shorthand: [Ar]$3d^5$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$; Shorthand: [Ar]$4s^2 3d^4$

Solution:
Neutral iron, Fe, has $Z = 26$ (26 electrons). Fe^{3+} has lost 3 electrons, so electrons remaining $= 26 - 3 = 23$. When removing electrons from transition metals remove $4s$ electrons before $3d$. Start from neutral Fe configuration: neutral Fe often written [Ar]$4s^2 3d^6$. Remove two $4s$ electrons first (-¿ $4s^0 3d^6$), then remove one $3d$ electron (-¿ $3d^5$). So the result is:
Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$ Check count: $2 + 2 + 6 + 2 + 6 + 5 = 23$.
Shorthand: [Ar]$3d^5$.

Answer: c. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$; [Ar]$3d^5$

Content (fully worked out problems & answers) continued...

(18) Write both the longhand and shorthand electron configurations for sulfide ion, S^{2-}.

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^4$; Shorthand: [Ne]$3s^2 3p^4$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6$; Shorthand: [Ar]

c. Longhand: $1s^2 2s^2 2p^6 3s^1 3p^6$; Shorthand: [Ne]$3s^1 3p^6$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^5$; Shorthand: [Ne]$3s^2 3p^5$

Solution:
Neutral sulfur S has $Z = 16$ (16 electrons). S^{2-} gains 2 electrons $\Rightarrow$ total electrons $= 16 + 2 = 18$. The configuration for 18 electrons is the same as argon. Longhand filling:

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$$

Check count: $2 + 2 + 6 + 2 + 6 = 18$.
Shorthand: the core is argon, so [Ar]. (One can also write [Ne]$3s^2 3p^6$ but [Ar] is simplest.)

Answer: b. $1s^2 2s^2 2p^6 3s^2 3p^6$; [Ar]

Content (fully worked out problems & answers) continued...

(19) Write both the longhand and shorthand electron configurations for neutral copper (Cu).

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$; Shorthand: $[\text{Ar}]3d^{10}4s^1$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^9$; Shorthand: $[\text{Ar}]4s^2 3d^9$

c. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^{11}$; Shorthand: $[\text{Ar}]3d^{11}$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^9$; Shorthand: $[\text{Ar}]4s^1 3d^9$

Solution:
Copper, Cu, $Z = 29$ so neutral Cu has 29 electrons. The observed (and most stable) configuration is an exception to the naive Aufbau filling: a filled d subshell ($3d^{10}$) plus a single $4s$ electron is favored. Thus longhand:

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$$

Check count: $2 + 2 + 6 + 2 + 6 + 10 + 1 = 29$.
Shorthand: core through 3p is argon, so $[\text{Ar}]3d^{10}4s^1$. Note option (a) orders as $3d^{10}4s^1$ but writing as $[\text{Ar}]4s^1 3d^{10}$ is also seen; the conventional shorthand groups the d after the noble gas core.

Answer: a. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$; $[\text{Ar}]3d^{10}4s^1$

Content (fully worked out problems & answers) continued...

(20) Write both the longhand and shorthand electron configurations for Mn^{2+}.

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$; Shorthand: [Ar]$3d^5$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$; Shorthand: [Ar]$4s^2 3d^3$

c. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$; Shorthand: [Ar]$3d^6$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^4$; Shorthand: [Ar]$4s^1 3d^4$

Solution:
Manganese, Mn, has $Z = 25$ (25 electrons) in the neutral atom. Mn^{2+} has lost 2 electrons $\Rightarrow$ electrons remaining $= 25 - 2 = 23$. Neutral Mn is commonly written [Ar]$4s^2 3d^5$. Remove the two $4s$ electrons first when forming cations from transition metals, leaving $4s^0 3d^5$. So the resulting configuration is:
Longhand: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$ Check: $2 + 2 + 6 + 2 + 6 + 5 = 23$.
Shorthand: [Ar]$3d^5$.

Answer: a. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$; [Ar]$3d^5$

Content (fully worked out problems & answers) continued...

(22) What is an electron orbital?

a. A circular path around the nucleus like planets orbiting the sun

b. A three-dimensional region where there is a high probability of finding an electron

c. A fixed energy value without any spatial meaning

d. A positively charged particle in the nucleus

Solution:
An orbital is not a fixed path (like planets), but rather a mathematical description of the region in space where an electron is most likely to be found. It is based on quantum mechanics and describes electron probability distributions.

Answer: b. A three-dimensional region where there is a high probability of finding an electron

Content (fully worked out problems & answers) continued...

(23) Which statement best describes the difference between an orbital and an orbit?

a. Both mean the same thing: a circular path around the nucleus

b. An orbit is a fixed circular path, while an orbital is a region of high probability for an electron

c. An orbital is smaller than an orbit

d. An orbit describes the nucleus, while an orbital describes protons

Solution:
Bohr's model described electrons in fixed circular paths called orbits. Modern quantum mechanics replaces this idea with orbitals, which are probability regions where electrons are most likely found. Orbitals are not fixed paths, but probability clouds.

Answer: b. An orbit is a fixed circular path, while an orbital is a region of high probability for an electron

Content (fully worked out problems & answers) continued...

(24) Draw the electron diagram (orbital boxes) for neutral oxygen (O). Place arrows (↑, ↓) to indicate electrons.

Empty orbital diagram (TikZ)

3p	☐	☐	☐
3s	☐		
2p	☐	☐	☐
2s	☐		
1s	☐		

Solution: Oxygen has 8 electrons. Filling orbitals using Aufbau/Hund/Pauli:

2p	↑	↑	↑
2s	↑↓		
1s	↑↓		

Answer: $1s^2\ 2s^2\ 2p^4$; 2 unpaired electrons in 2p orbitals.

Content (fully worked out problems & answers) continued...

(25) Draw the electron diagram (orbital boxes) for the chloride ion, Cl^-. Place arrows (↑, ↓) to indicate electrons.

Empty orbital diagram (TikZ)

3p	☐	☐	☐
3s	☐		
2p	☐	☐	☐
2s	☐		
1s	☐		

Solution: Electron configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. All orbitals fully filled:

3p	↑↓	↑↓	↑↓
3s	↑↓		
2p	↑↓	↑↓	↑↓
2s	↑↓		
1s	↑↓		

Answer: Fully filled orbitals; closed shell.

Content (fully worked out problems & answers) continued...

(26) Draw the electron diagram (orbital boxes) for neutral iron (Fe). Place arrows (↑, ↓) to indicate electrons. Use Aufbau, Hund, and Pauli. Pay attention to how 4s and 3d are filled and later how electrons are paired/unpaired in 3d.

Empty orbital diagram (TikZ)

3d	☐	☐	☐	☐	☐
4s	☐				
3p	☐	☐	☐		
3s	☐				
2p	☐	☐	☐		
2s	☐				
1s	☐				

Solution: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^6$:

3d	↑	↑	↑	↑	↑↓
4s	↑↓				
3p	↑↓	↑↓	↑↓		
3s	↑↓				
2p	↑↓	↑↓	↑↓		
2s	↑↓				
1s	↑↓				

Answer: 4 unpaired electrons in 3d.

Content (fully worked out problems & answers) continued...

(27) Draw the electron diagram (orbital boxes) for neutral copper (Cu). Place arrows (↑, ↓) to indicate electrons.

Empty orbital diagram (TikZ)

3d	☐ ☐ ☐ ☐ ☐
4s	☐
3p	☐ ☐ ☐
3s	☐
2p	☐ ☐ ☐
2s	☐
1s	☐

Solution: Cu has electron configuration [Ar] $3d^{10}$ $4s^{1}$:

3d	↑↓ ↑↓ ↑↓ ↑↓ ↑↓
4s	↑
3p	↑↓ ↑↓ ↑↓
3s	↑↓
2p	↑↓ ↑↓ ↑↓
2s	↑↓
1s	↑↓

Answer: $4s^{1}$ $3d^{10}$; 1 unpaired electron in 4s.

Content (fully worked out problems & answers) continued...

(28) Draw the electron diagram (orbital boxes) for the sulfide ion, S^{2-}. Place arrows (↑, ↓) to indicate electrons. Place arrows consistent with Aufbau, Hund, and Pauli.

Empty orbital diagram (TikZ)

3p	☐	☐	☐	
3s	☐			
2p	☐	☐	☐	Fill: 18 electrons; closed shell is expected.
2s	☐			
1s	☐			

Solution: Electron configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ (like Ar):

3p	↑↓	↑↓	↑↓
3s	↑↓		
2p	↑↓	↑↓	↑↓
2s	↑↓		
1s	↑↓		

Answer: All orbitals filled; closed-shell configuration.

Content (fully worked out problems & answers) continued...

(29) A tectonic plate moves 860 km per day. What is its speed in m/s?

a. 0.995 m/s

b. 9.95 m/s

c. 95.9 m/s

d. 0.0959 m/s

Answer:

$$\text{Speed} = \frac{860\ \cancel{\text{km}}}{\cancel{\text{day}}} \times \frac{1000\ \text{m}}{1\ \cancel{\text{km}}} \times \frac{1\ \cancel{\text{day}}}{86400\ \text{s}} = 9.95\ \text{m/s}$$

Content (fully worked out problems & answers) continued...

(30) Convert 750 mL to in^3. (Use 1 in = 2.54 cm)

a. 4.58 in^3

b. 45.8 in^3

c. 458 in^3

d. 0.458 in^3

Answer:

$$750 \text{ mL} = 750 \text{ cm}^3, \quad 1 \text{ in}^3 = (2.54 \text{ cm})^3 = 16.387 \text{ cm}^3$$

$$\text{Volume} = \frac{750 \text{ cm}^3}{16.387 \text{ cm}^3/\text{in}^3} = 45.8 \text{ in}^3$$

Content (fully worked out problems & answers) continued...

(31) Convert 88.0 kPa to atm. (Use 1 atm = 101.325 kPa)

a. 0.868 atm

b. 0.880 atm

c. 1.15 atm

d. 0.815 atm

Answer:

$$\text{Pressure} = \frac{88.0\ \cancel{\text{kPa}}}{1} \times \frac{1\ \text{atm}}{101.325\ \cancel{\text{kPa}}} = 0.868\ \text{atm}$$

Content (fully worked out problems & answers) continued...

(32) Convert 4.6 g/L to mg/dL.

a. 0.46 mg/dL

b. 4.6 mg/dL

c. 46 mg/dL

d. 460 mg/dL

Answer:

$$\frac{4.6\ \cancel{g}}{\cancel{L}} \times \frac{1000\ \text{mg}}{1\ \cancel{g}} \times \frac{1\ \text{dL}}{10\ \cancel{L}} = 46\ \text{mg/dL}$$

Content (fully worked out problems & answers) continued...

(33) A block has mass 245.0 g and dimensions 4.50 cm × 3.20 cm × 2.10 cm. Find its density in kg/m^3.

a. 8.10×10^2 kg/m^3

b. 8.10×10^3 kg/m^3

c. 8.10×10^4 kg/m^3

d. 8.10×10^5 kg/m^3

Answer:

$$V = (4.50)(3.20)(2.10)\ \text{cm}^3 = 30.2\ \text{cm}^3$$

$$\rho = \frac{245.0\ \text{g}}{30.2\ \text{cm}^3} = 8.10\ \text{g/mL} = 8.10\ \frac{\text{g}}{\text{cm}^3}$$

$$\text{Density} = \left(8.10\ \frac{\text{g}}{\text{cm}^3}\right) \times \frac{1\ \text{kg}}{1000\ \text{g}} \times \frac{10^6\ \text{cm}^3}{1\ \text{m}^3} = 8.10 \times 10^3\ \text{kg/m}^3$$

Content (fully worked out problems & answers) continued...

(34) A 0.500 g sample of NaCl dissolves in 250.0 mL of water. What is the concentration in g/L?

a. 0.125 g/L

b. 2.00 g/L

c. 1.25 g/L

d. 0.250 g/L

Answer:

$$\text{Concentration} = \frac{0.500 \text{ g}}{0.250 \text{ L}} = 2.00 \text{ g/L}$$

Content (fully worked out problems & answers) continued...

(35) Convert 987.0 mbar to Pa.

a. 9.870 Pa

b. 9.87×10^2 Pa

c. 9.87×10^4 Pa

d. 9.87×10^5 Pa

Answer:

$$987.0 \text{ mbar} \times \frac{100 \text{ Pa}}{1 \text{ mbar}} = 9.870 \times 10^4 \text{ Pa}$$

Content (fully worked out problems & answers) continued...

(36) A sample occupies 250.0 mL and has a mass of 0.890 g. Calculate its density.

a. 3.56×10^{-3} g/mL

b. 0.890 g/mL

c. 3.56 g/mL

d. 35.6 g/mL

Answer:

$$\rho = \frac{0.890 \text{ g}}{250.0 \text{ mL}} = 3.56 \times 10^{-3} \text{ g/mL}$$

Content (fully worked out problems & answers) continued...

(37) A beaker contains 3.50 L of solution. Express this volume in cubic decimeters (dm^3) using unit factors.

a. 0.00350 dm^3

b. 3.50 dm^3

c. 35.0 dm^3

d. 350 dm^3

Answer:

$$3.50 \text{ L} \times \frac{10^{-3} \text{ dm}^3}{1 \text{ L}} = 0.00350 \text{ dm}^3$$

Content (fully worked out problems & answers) continued...

(38) A student measures 0.00250 kg of a substance. Express this in grams.

a. 0.00250 g

b. 0.0250 g

c. 2.50 g

d. 250 g

Answer:

$$0.00250 \text{ kg} \times \frac{1000 \text{ g}}{1 \text{ kg}} = 2.50 \text{ g}$$

Content (fully worked out problems & answers) continued...

(39) A 5.00 mL sample of water weighs 4.98 g. What is its density?

a. 0.996 g/mL

b. 1.00 g/mL

c. 1.02 g/mL

d. 1.10 g/mL

Answer:

$$\rho = \frac{4.98 \text{ g}}{5.00 \text{ mL}} = 0.996 \text{ g/mL}$$

Content (fully worked out problems & answers) continued...

(40) Which of the following is an isotope of carbon?

a. ^{12}C

b. ^{14}C

c. Both a and b

d. ^{16}O

Solution:
Isotopes are atoms of the same element with different mass numbers (different neutrons). Both ^{12}C and ^{14}C are carbon isotopes.

Answer: c. Both a and b

Content (fully worked out problems & answers) continued...

(41) Which subatomic particle determines the identity of an element?

a. Proton

b. Electron

c. Neutron

d. Nucleus

Solution:
The number of protons (atomic number) defines the element. Changing the number of protons changes the element.

Answer: a. Proton

Content (fully worked out problems & answers) continued...

(42) An atom of sodium (Na) loses one electron. What is the resulting ion?

a. Na^-

b. Na^+

c. Na^{2+}

d. Neutral Na

Solution:
Losing one electron creates a positive charge. Sodium forms Na^+ when it loses one electron.

Answer: b. Na^+

Content (fully worked out problems & answers) continued...

(43) Which of the following are isotopes of the same element?

a. ^{1}H and ^{2}H

b. ^{12}C and ^{14}N

c. ^{16}O and ^{18}F

d. ^{23}Na and ^{24}Mg

Solution:
Isotopes are atoms with the same number of protons but different neutrons. ^{1}H and ^{2}H are hydrogen isotopes.

Answer: a. ^{1}H and ^{2}H

Content (fully worked out problems & answers) continued...

(46) Which type of bond is expected between Na and Cl in NaCl?

a. Nonpolar covalent

b. Polar covalent

c. Ionic

d. Metallic

Solution:
Electronegativity difference between Na (0.9) and Cl (3.0) is about 2.1, which is large. A difference greater than ~ 2.0 typically indicates ionic bonding. Thus, Na transfers an electron to Cl, forming an ionic bond.

Answer: c. Ionic

Content (fully worked out problems & answers) continued...

(47) Which of the following has the highest electronegativity?

a. Lithium

b. Carbon

c. Fluorine

d. Oxygen

Solution:
Electronegativity increases across a period and decreases down a group. Fluorine is the most electronegative element (value ≈ 4.0), higher than oxygen (3.5), carbon (2.5), and lithium (1.0).

Answer: c. Fluorine

Content (fully worked out problems answers) continued...

(48) What is the correct name for MgF_2?

a. Magnesium difluoride

b. Magnesium fluoride

c. Magnesium(II) fluoride

d. Monomagnesium fluoride

Solution:
Magnesium is Group 2 (+2). Fluoride is -1, two are needed to balance. The correct name is magnesium fluoride.

Answer: b. Magnesium fluoride

Content (fully worked out problems & answers) continued...

(49) Which is the correct name for Cu_2O?

a. Copper(I) oxide

b. Copper(II) oxide

c. Dicopper monoxide

d. Copper oxide

Solution:
Oxygen is -2. Two Cu atoms balance it, so each Cu is +1. Correct name: Copper(I) oxide.

Answer: a. Copper(I) oxide

Content (fully worked out problems & answers) continued...

(50) What is the formula for lithium sulfide?

a. LiS

b. Li_2S

c. LiS_2

d. Li_2S_3

Solution:
Lithium is +1, sulfide is -2. Two Li^+ are required for one S^{2-}. Formula: Li_2S.

Answer: b. Li_2S

Content (fully worked out problems & answers) continued...

(51) What is the correct name for PbO_2?

a. Lead(II) oxide

b. Lead(IV) oxide

c. Lead oxide

d. Dilead dioxide

Solution:
Oxygen is -2, two atoms total -4. Pb must be +4. Correct name: Lead(IV) oxide.

Answer: b. Lead(IV) oxide

Content (fully worked out problems & answers) continued...

(52) Which of the following is the correct formula for strontium chloride?

a. SrCl

b. Sr_2Cl

c. $SrCl_2$

d. Sr_2Cl_3

Solution:
Strontium is Group 2 (+2). Chloride is -1, so two Cl^- are needed. Formula: $SrCl_2$.

Answer: c. $SrCl_2$

Content (fully worked out problems answers) continued...

(53) What is the correct name for $FeCl_3$?

a. Iron chloride

b. Iron(II) chloride

c. Iron(III) chloride

d. Ferrous chloride

Solution:
Each Cl is -1, three Cl give -3 total, so Fe must be +3. Use Roman numeral: Iron(III) chloride.

Answer: c. Iron(III) chloride

Content (fully worked out problems & answers) continued...

(54) Which is the correct name for CoO?

a. Cobalt oxide

b. Cobalt(II) oxide

c. Cobalt(III) oxide

d. Dicobalt monoxide

Solution:
O is -2, one Co balances with +2. That gives cobalt in the +2 oxidation state: Cobalt(II) oxide.

Answer: b. Cobalt(II) oxide

Content (fully worked out problems & answers) continued...

(55) What is the correct name for Cu_2S?

a. Copper sulfide

b. Copper(II) sulfide

c. Copper(I) sulfide

d. Dicopper monosulfide

Solution:
Sulfide is -2. Two Cu atoms give +2 total so each Cu is +1. Name uses Roman numeral: Copper(I) sulfide.

Answer: c. Copper(I) sulfide

Content (fully worked out problems answers) continued...

(56) What is the correct name for $Zn(NO_3)_2$?

a. Zinc nitrate

b. Zinc dinitrate

c. Zinc(II) nitrite

d. Zinc nitrite

Solution: Nitrate is -1; two nitrates give -2, so Zn must be $+2$. Name: zinc (common oxidation state $+2$) with nitrate.

Answer: a. Zinc nitrate

Content (fully worked out problems answers) continued...

(57) Which is the correct formula for silver sulfate?

a. $AgSO_4$

b. Ag_2SO_4

c. $Ag(SO_4)_2$

d. $Ag_2(SO_3)$

Solution: Sulfate is -2 and Ag is $+1$, so two Ag^+ are needed per sulfate: formula Ag_2SO_4.

Answer: b. Ag_2SO_4

(58) What is the correct name for $Cd(OH)_2$?

a. Cadmium hydroxide

b. Cadmium(II) hydroxide

c. Cadmium dihydroxide

d. Cadmium(I) hydroxide

Solution: Hydroxide is -1; two OH^- give -2, so Cd is $+2$. Proper systematic name: cadmium(II) hydroxide (commonly "cadmium hydroxide").

Answer: b. Cadmium(II) hydroxide

Content (fully worked out problems answers) continued...

(59) Which is the correct name for $Ga_2(CO_3)_3$?

a. Gallium carbonate

b. Gallium(II) carbonate

c. Gallium(III) carbonate

d. Digallium tricarbonate

Solution: Carbonate is -2; three give -6, two Ga supply $+6$ total so each Ga is $+3$. Name: gallium(III) carbonate (commonly "gallium carbonate").

Answer: c. Gallium(III) carbonate

Content (fully worked out problems answers) continued...

(60) What is the formula for ammonium phosphate?

a. $(NH_4)PO_4$

b. $(NH_4)_2PO_4$

c. $(NH_4)_3PO_4$

d. NH_4PO_4

Solution: Phosphate is -3; ammonium is $+1$ so three NH_4^+ are needed to balance one PO_4^{3-}: $(NH_4)_3PO_4$.

Answer: c. $(NH_4)_3PO_4$

Content (fully worked out problems answers) continued...

(61) What is the correct name for $Al_2(SO_4)_3$?

a. Aluminum sulfate

b. Aluminum(III) sulfate

c. Dialuminum trisulfate

d. Aluminum sulfite

Solution: Sulfate is -2; three sulfates give -6, two Al give $+6$ so Al is $+3$. Common name: aluminum sulfate (aluminum(III) sulfate is equivalent).

Answer: a. Aluminum sulfate

Content (fully worked out problems answers) continued...

(62) Which is the correct formula for zinc acetate?

a. $Zn(C_2H_3O_2)$

b. $Zn(C_2H_3O_2)_2$

c. $Zn_2(C_2H_3O_2)_3$

d. $Zn_2(C_2H_3O_2)_2$

Solution: Acetate is -1; Zn is $+2$, so two acetates are needed: $Zn(C_2H_3O_2)_2$.

Answer: b. $Zn(C_2H_3O_2)_2$

(63) What is the correct name for $Ag_3(PO_4)$?

a. Silver phosphate

b. Silver(III) phosphate

c. Trisilver phosphate

d. Silver(I) phosphate

Solution: Phosphate is -3, Ag is $+1$ so three Ag^+ balance one PO_4^{3-}. Name: silver phosphate (silver(I) phosphate is equivalent).

Answer: a. Silver phosphate

Content (fully worked out problems answers) continued...

(64) Which is the correct formula for cadmium sulfite?

a. $CdSO_3$

b. $Cd_2(SO_3)_3$

c. $Cd(SO_3)_2$

d. $Cd_3(SO_3)_2$

Solution: Sulfite is -2; Cd is typically $+2$, so a 1:1 formula $CdSO_3$.

Answer: a. $CdSO_3$

Content (fully worked out problems answers) continued...

(65) What is the correct name for $Ga(NO_2)_3$?

a. Gallium nitrate

b. Gallium nitrite

c. Gallium(III) nitrite

d. Gallium trinitrate

Solution: Nitrite is -1; three nitrites give -3, Ga must be $+3$. Proper name: gallium(III) nitrite (commonly "gallium nitrite").

Answer: c. Gallium(III) nitrite

Content (fully worked out problems answers) continued...

(66) What is the formula for ammonium chromate?

a. $(NH_4)CrO_4$

b. $(NH_4)_2CrO_4$

c. $(NH_4)_3CrO_4$

d. NH_4CrO_4

Solution: Chromate is CrO_4^{2-}; ammonium is +1, so two ammoniums are required: $(NH_4)_2CrO_4$.

Answer: b. $(NH_4)_2CrO_4$

Content (fully worked out problems answers) continued...

(67) Which is the correct name for $Zn(CN)_2$?

a. Zinc cyanide

b. Zinc dicyanide

c. Zinc(II) cyanide

d. Zinc cyanate

Solution: Cyanide is -1; two CN^- balance Zn^{2+} giving $Zn(CN)_2$. Name: zinc cyanide (or zinc(II) cyanide).

Answer: a. Zinc cyanide

Content (fully worked out problems answers) continued...

(68) What is the correct name for $Al(OH)_3$?

a. Aluminum hydroxide

b. Aluminum(III) hydroxide

c. Trialuminum hydroxide

d. Aluminum dihydroxide

Solution: Hydroxide is -1; three OH^- give -3, so Al is $+3$. Name: aluminum hydroxide (aluminum(III) hydroxide is explicit).

Answer: a. Aluminum hydroxide

Content (fully worked out problems answers) continued...

(69) Which is the correct formula for silver acetate?

a. $AgC_2H_3O_2$

b. $Ag(C_2H_3O_2)_2$

c. $Ag_2(C_2H_3O_2)$

d. $Ag_2(C_2H_3O_2)_2$

Solution: Acetate is singly charged (-1) and Ag is $+1$, so one acetate per Ag: $AgC_2H_3O_2$.

Answer: a. $AgC_2H_3O_2$

Content (fully worked out problems answers) continued...

(70) What is the correct name for $Cd_3(PO_4)_2$?

a. Cadmium phosphate

b. Cadmium(III) phosphate

c. Tricadmium diphosphate

d. Cadmium(II) phosphate

Solution: Phosphate is -3, two phosphates give -6, three Cd give $+6$, so each Cd is $+2$. Name: cadmium(II) phosphate (commonly "cadmium phosphate").

Answer: d. Cadmium(II) phosphate

Content (fully worked out problems answers) continued...

(71) Which is the correct formula for gallium hydroxide?

a. GaOH

b. $Ga(OH)_2$

c. $Ga(OH)_3$

d. $Ga_2(OH)_3$

Solution: Ga is +3 and hydroxide is −1, so three OH^- required: $Ga(OH)_3$.

Answer: c. $Ga(OH)_3$

Content (fully worked out problems answers) continued...

(72) What is the correct name for $ZnSO_3$?

a. Zinc sulfate

b. Zinc sulfite

c. Zinc(II) sulfite

d. Zinc sulfur trioxide

Solution: Sulfite is SO_3^{2-} and Zn is +2, so formula $ZnSO_3$ is 1:1. Name: zinc sulfite (systematically zinc(II) sulfite).

Answer: c. Zinc(II) sulfite

Content (fully worked out problems answers) continued...

(73) Which is the correct formula for ammonium nitrate?

a. $(NH_4)NO_2$

b. $(NH_4)NO_3$

c. $(NH_4)_2NO_3$

d. NH_4NO_2

Solution: Nitrate is NO_3^- and ammonium is NH_4^+; they combine 1:1 giving $(NH_4)NO_3$ (often written NH_4NO_3).

Answer: b. $(NH_4)NO_3$

Content (fully worked out problems answers)continued...

(74) What is the correct name for Ag_2CO_3?

a. Silver carbonate

b. Silver(II) carbonate

c. Disilver carbonate

d. Silver(I) carbonate

Solution: Carbonate is -2; Ag is $+1$, so two Ag^+ balance CO_3^{2-}: Ag_2CO_3. Name: silver carbonate (silver(I) carbonate is equivalent).

Answer: a. Silver carbonate

Content (fully worked out problems answers) continued...

(75) Which is the correct formula for aluminum nitrite?

a. $Al(NO_2)$

b. $Al(NO_2)_2$

c. $Al(NO_2)_3$

d. $Al_2(NO_2)_3$

Solution: Nitrite is NO_2^-; Al is +3, so three nitrite ions are needed: $Al(NO_2)_3$.

Answer: c. $Al(NO_2)_3$

Content (fully worked out problems & answers) continued...

(76) Start from the Lewis dot of a neutral sodium atom (Na) and draw the Lewis dot structure for the sodium ion, Na^+. Indicate what happens to the valence electron and the resulting charge.

Solution:
Neutral Na has 1 valence electron (one dot). Removing that electron gives Na^+ with no valence dots and a +1 charge.

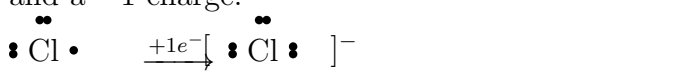
$$\text{Na}\bullet \quad \xrightarrow{-1e^-} [\ \text{Na}\ ]^+$$

Answer: Na^+ — electron removed, no valence dots, charge +1.

Content (fully worked out problems & answers) continued...

(77) Given the Lewis dot for a neutral chlorine atom, draw the Lewis dot structure for the chloride ion, Cl^-. Show where the added electron is and indicate the charge.

Solution:
Neutral Cl has 7 valence electrons (7 dots). Adding one electron gives 8 (octet) and a -1 charge.

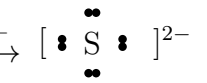


Answer: Cl^- — octet completed, charge -1.

Content (fully worked out problems & answers) continued...

(78) Start from the Lewis dot of a neutral sulfur atom (S) and draw the Lewis dot structure for the sulfide ion, S^{2-}. Show electron counting and charge.

Solution:
Neutral S has 6 valence electrons. Adding 2 electrons yields 8 (octet) and a -2 charge.



Answer: S^{2-} — octet, charge -2.

Content (fully worked out problems & answers) continued...

(79) Given the Lewis dot of a neutral magnesium atom (Mg), draw the Lewis structure for Mg^{2+}. Indicate removal of valence electrons and the resulting charge.

Solution:
Neutral Mg has 2 valence electrons (two dots). Removing both gives Mg^{2+} with no valence dots and a +2 charge.

$$\text{Mg}\,\bullet\bullet \xrightarrow{-2e^-} [\ \text{Mg}\ ]^{2+}$$

Answer: Mg^{2+} — two electrons removed, charge +2.

Content (fully worked out problems & answers) continued...

(80) Start from the Lewis dot for a neutral copper atom (showing its single 4s valence electron as one dot) and draw the Lewis structure for the cuprous ion, Cu^+. Explain briefly.

Solution:
A common simple Lewis-dot convention places one valence dot for Cu ($4s^1$). Removing that electron gives Cu^+ with no valence dot and charge +1. (Note: d-electrons are not shown in simple Lewis-dot for many transition metals.)

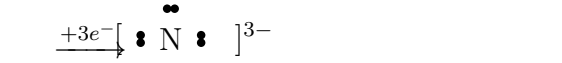
$$\mathrm{Cu}\bullet \quad \xrightarrow{-1e^-} \quad [\,\mathrm{Cu}\,]^+$$

Answer: Cu^+ — 4s electron removed, no valence dot, charge +1.

Content (fully worked out problems & answers) continued...

(81) Given the Lewis dot for neutral nitrogen (N), draw the Lewis dot for nitride ion, N^{3-}, and indicate electron counting and charge.

Solution:
Neutral N has 5 valence electrons (5 dots). Adding 3 electrons yields 8 (octet) and a -3 charge.



Answer: N^{3-} — octet filled, charge -3.

Content (fully worked out problems answers) continued...

(82) Draw the Lewis dot structure for oxygen (O).

Solution: Oxygen is in group 16, so it has 6 valence electrons. Place them as 2 lone pairs and 2 single dots around O.

Answer:

$$\cdot\, O \cdot :$$

(two lone pairs, two single electrons)

Content (fully worked out problems answers) continued...

(83) Draw the Lewis dot structure for carbon (C).

Solution: Carbon is in group 14, so it has 4 valence electrons. Place one dot on each side of C.

Answer: One dot on each side of C.

Content (fully worked out problems answers) continued...

(84) Draw the Lewis structure of H_2.

Solution: Each H has 1 valence electron. They share a pair, forming a single bond.

Answer: H–H

Content (fully worked out problems answers) continued...

(85) Draw the Lewis structure for H_2O.

Solution: Oxygen has 6 valence electrons, hydrogen has 1 each. O bonds with two H, and has two lone pairs.

Answer: H–O–H with 2 lone pairs on O.

Content (fully worked out problems answers) continued...

(86) Draw the Lewis structure of CO_2.

Solution: Carbon has 4 valence electrons, oxygen 6 each. To satisfy octets, C forms double bonds to each O.

Answer: O=C=O with two lone pairs on each O.

Content (fully worked out problems answers) continued...

(87) Draw the Lewis structure for NH_3.

Solution: N has 5 valence electrons, H has 1 each. Three N–H bonds plus one lone pair on N.

Answer: N in center, 3 H attached, 1 lone pair on N.

Content (fully worked out problems answers) continued...

(88) Draw the Lewis structure of CH_4.

Solution: C has 4 valence electrons, H has 1 each. Four C–H single bonds. No lone pairs on C.

Answer: Central C with 4 H bonded.

Content (fully worked out problems answers) continued...

(89) Draw the Lewis structure for O_3.

Solution: 18 total valence electrons. Resonance: one double bond and one single bond between O atoms, with lone pairs.

Answer: Resonance structures: O=O–O O–O=O with lone pairs.

Content (fully worked out problems answers) continued...

(90) Draw the Lewis structure of SO_2.

Solution: 18 valence electrons. Resonance: one double bond S=O and one single bond S–O with lone pairs. Expanded octet possible.

Answer: Resonance between S=O–O and O–S=O.

Content (fully worked out problems answers) continued...

(91) Draw the Lewis structure of NO_3^-.

Solution: 24 valence electrons. N bonded to 3 O, one double bond, two single bonds. Negative charge delocalized $\rightarrow$ resonance.

Answer: Resonance: N–O double bond rotates among the 3 O atoms.

Content (fully worked out problems answers) continued...

(92) Draw the Lewis structure of SF_6.

Solution: S has 6 valence electrons, each F has 7. S forms 6 single bonds with F, using expanded octet (12 electrons on S).

Answer: Central S with 6 F atoms, each F with 3 lone pairs.

Content (fully worked out problems answers) continued...

(93) Draw the Lewis structure of XeF_4.

Solution: Xe has 8 valence electrons, F has 7 each. Xe forms 4 bonds to F, with 2 lone pairs remaining on Xe (expanded octet).

Answer: Central Xe with 4 F atoms, each F has 3 lone pairs, Xe has 2 lone pairs.

(94) Draw the electron diagram (orbital boxes) for neutral neon (Ne). Place arrows (↑, ↓) to indicate electrons.

Electron configuration diagram

4p	☐	☐	☐		
3d	☐	☐	☐	☐	☐
4s	☐				
3p	☐	☐	☐		
3s	☐				
2p	☐	☐	☐		
2s	☐				
1s	☐				

Electron configuration diagram

4p	☐	☐	☐		
3d	☐	☐	☐	☐	☐
4s	☐				
3p	☐	☐	☐		
3s	☐				
2p	↑↓	↑↓	↑↓		
2s	↑↓				
1s	↑↓				

(95) Draw the electron diagram (orbital boxes) for neutral magnesium (Mg). Place arrows (↑, ↓) to indicate electrons.

Electron configuration diagram

4p	☐ ☐ ☐
3d	☐ ☐ ☐ ☐ ☐
4s	☐
3p	☐ ☐ ☐
3s	☐
2p	☐ ☐ ☐
2s	☐
1s	☐

Electron configuration diagram

4p	☐ ☐ ☐
3d	☐ ☐ ☐ ☐ ☐
4s	☐
3p	☐ ☐ ☐
3s	↑↓
2p	↑↓ ↑↓ ↑↓
2s	↑↓
1s	↑↓

(96) Draw the electron diagram (orbital boxes) for neutral phosphorus (P). Place arrows (↑, ↓) to indicate electrons.

Electron configuration diagram

4p			
3d			
4s			
3p			
3s			
2p			
2s			
1s			

Electron configuration diagram

4p			
3d			
4s			
3p	↑	↑	↑
3s	↑↓		
2p	↑↓	↑↓	↑↓
2s	↑↓		
1s	↑↓		

(97) Draw the electron diagram (orbital boxes) for neutral sulfur (S). Place arrows (↑, ↓) to indicate electrons.

Electron configuration diagram

4p					
3d					
4s					
3p					
3s					
2p					
2s					
1s					

Electron configuration diagram

4p					
3d					
4s					
3p	↑↓	↑	↑		
3s	↑↓				
2p	↑↓	↑↓	↑↓		
2s	↑↓				
1s	↑↓				

(98) Draw the electron diagram (orbital boxes) for neutral chlorine (Cl). Place arrows (↑, ↓) to indicate electrons.

Electron configuration diagram

4p					
3d					
4s					
3p					
3s					
2p					
2s					
1s					

Electron configuration diagram

4p					
3d					
4s					
3p	↑↓	↑↓	↑		
3s	↑↓				
2p	↑↓	↑↓	↑↓		
2s	↑↓				
1s	↑↓				

(99) Draw the electron diagram (orbital boxes) for neutral calcium (Ca). Place arrows (↑, ↓) to indicate electrons.

Electron configuration diagram

4p	☐ ☐ ☐
3d	☐ ☐ ☐ ☐ ☐
4s	☐
3p	☐ ☐ ☐
3s	☐
2p	☐ ☐ ☐
2s	☐
1s	☐

Electron configuration diagram

4p	☐ ☐ ☐
3d	☐ ☐ ☐ ☐ ☐
4s	↑↓
3p	↑↓ ↑↓ ↑↓
3s	↑↓
2p	↑↓ ↑↓ ↑↓
2s	↑↓
1s	↑↓

(100) Draw the electron diagram (orbital boxes) for neutral titanium (Ti). Place arrows (↑, ↓) to indicate electrons.

Electron configuration diagram

4p	☐ ☐ ☐
3d	☐ ☐ ☐ ☐ ☐
4s	☐
3p	☐ ☐ ☐
3s	☐
2p	☐ ☐ ☐
2s	☐
1s	☐

Electron configuration diagram

4p	☐ ☐ ☐
3d	↑ ↑ ☐ ☐ ☐
4s	↑↓
3p	↑↓ ↑↓ ↑↓
3s	↑↓
2p	↑↓ ↑↓ ↑↓
2s	↑↓
1s	↑↓

(101) Draw the electron diagram (orbital boxes) for neutral vanadium (V). Place arrows (↑, ↓) to indicate electrons.

Electron configuration diagram

4p	☐	☐	☐
3d	☐	☐	☐ ☐ ☐
4s	☐		
3p	☐	☐	☐
3s	☐		
2p	☐	☐	☐
2s	☐		
1s	☐		

Electron configuration diagram

4p	☐	☐	☐
3d	↑	↑	↑ ☐ ☐
4s	↑↓		
3p	↑↓	↑↓	↑↓
3s	↑↓		
2p	↑↓	↑↓	↑↓
2s	↑↓		
1s	↑↓		

Content (fully worked out problems answers) continued...

(102) Draw the electron diagram (orbital boxes) for neutral zinc (Zn). Place arrows (↑, ↓) to indicate electrons.

Electron configuration diagram

4p	☐ ☐ ☐
3d	☐ ☐ ☐ ☐ ☐
4s	☐
3p	☐ ☐ ☐
3s	☐
2p	☐ ☐ ☐
2s	☐
1s	☐

Electron configuration diagram

4p	☐ ☐ ☐
3d	↑↓ ↑↓ ↑↓ ↑↓ ↑↓
4s	↑↓
3p	↑↓ ↑↓ ↑↓
3s	↑↓
2p	↑↓ ↑↓ ↑↓
2s	↑↓
1s	↑↓

Checking for Understanding (general conceptual questions)

1. How do you identify cations and anions, and what processes form them?
2. How can electron configurations predict the chemical behavior and reactivity of elements?
3. What is the difference between an atom, an ion, and an isotope, and how do these differences affect chemical properties?
4. How do orbitals differ from orbits, and why is this distinction important for understanding electron arrangements?
5. How can you determine the shape of a molecule using VSEPR theory?
6. What is the difference between ionic, polar covalent, and nonpolar covalent bonds, and how can you predict which will form?
7. How do electronegativity differences relate to bond polarity and molecular polarity?
8. How do you determine if a molecule is polar or nonpolar overall?
9. How do the number of protons, neutrons, and electrons determine the identity and charge of an atom or ion?
10. How can you organize elements using the periodic table, and what patterns help predict properties?
11. How does atomic and molecular structure influence physical properties such as density or polarity?
12. How do measurements in chemistry, such as mass, volume, and concentration, relate to chemical calculations and formulas?
13. How do isotopes of the same element differ in mass but retain the same chemical behavior?
14. How do specific orbitals (s, p, d, f) relate to electron distribution in atoms and the periodic trends?
15. How do scientific notation and powers of ten help when representing very large or very small quantities in chemistry?

Activity 2

Collaborative Learning Techniques	Informal Boardwork Tutoring
Learning Strategy	Student-Driven Boardwork Model (modified)
Time	12:30—12:50
Materials	Colored whiteboard markers, large whiteboard space (divided into sections), optional reference charts (periodic table, orbital diagrams, polyatomic ion list).
Lecture/Textbook/Old Plans/Slide References	Chapter 3: Periodic Table, Chapter 4: Chemical Bonding: The Ionic Bond Model, Chapter 5: The Chemical Bonding: The Covalent Bond Model

Definitions

1. **Jigsaw:** Jigsaws make groups interdependent by splitting a large topic into sub-parts, where each group becomes an "expert" on one piece before teaching it back to the larger group.
2. **Boardwork Model (modified):** Instead of splitting a prepared problem into pieces, this free-flowing version invites students to volunteer questions they want clarified (conceptual or problem-based). Students or small groups attempt answers on the whiteboard, narrating their reasoning while peers observe, critique, and contribute. The board is partitioned so multiple questions can be worked on simultaneously. The focus is on interactive explanation and collaborative correction.

Instructions (Detailed Activity Breakdown)

Objective: Create an open, collaborative environment where students bring forward their own chemistry questions (from notes, homework, or general concepts), work through them on the board, and refine their understanding with peer and facilitator input.

Setup:

- Divide the whiteboard into 3–4 sections to allow simultaneous work.
- Provide markers of different colors so students can distinguish steps, corrections, or alternative approaches.
- Encourage students to bring forward any questions — broad conceptual ones (e.g., "Why does 4s fill before 3d?") or specific problems they struggled with in homework.

Running the Activity:

1. Invite students to propose questions they would like answered; write them at the top of each board section.
2. Call on volunteers (or pairs) to attempt answering the question at the board, narrating their thought process.
3. Pause after each major step for peer input — others can add corrections, clarifications, or alternative reasoning.
4. Rotate students so that new people get to attempt the next question, or continue working on unresolved steps.
5. If a question stalls, the facilitator provides guiding prompts rather than the full solution immediately.
6. After each solution, briefly summarize the key concept and ensure the group agrees on the corrected/complete reasoning.
7. Continue until all student-proposed questions are addressed or time runs out.

Facilitator Role:

- Prompt students to supply the questions rather than relying on pre-written problems.
- Ensure participation is spread across the group — rotate boardwork between students.
- Ask probing questions ("Why did you choose that orbital?" "How do you know the molecule is polar?") to deepen reasoning.
- Step in only when necessary to correct persistent misconceptions, but allow peer-to-peer correction first.
- At the end, recap common themes and highlight problem-solving strategies students can carry into independent study.

Checking for Understanding (questions & answers)

Questions:

1. How does letting students bring their own questions encourage them to focus on areas where they feel the least confident?
2. Why is it helpful for peers to see how someone else sets up or explains a problem on the board?
3. In what ways does this approach encourage students to reflect on what they understand versus what they need clarified?
4. How can having peers correct or expand an explanation prevent misunderstandings from persisting?
5. How does this model balance independent thinking with group collaboration?
6. What strategies can students use to decide whether a question is conceptual (big picture) or procedural (step-by-step)?
7. How does moving between different student questions help the group connect broader concepts with specific details?
8. How can presenting reasoning out loud improve a student's ability to communicate chemical ideas clearly?
9. What advantages do students gain from watching multiple approaches to the same question be worked out at the board?
10. How does this activity help students apply theoretical knowledge to real problem-solving situations?
11. In what ways can the instructor or SI leader use the questions students ask to identify where the class struggles most?
12. How does working through student-generated questions in real time strengthen long-term understanding of chemistry concepts?

Closer

Collaborative Learning Techniques	Group Survey
Learning Strategy	Brain dump
Time	12:50—12:55
Materials	Index cards, writing utensils
Lecture/Textbook/Old Plans/Slide References	Chapter 3: Periodic Table, Chapter 4: Chemical Bonding: The Ionic Bond Model, Chapter 5: The Chemical Bonding: The Covalent Bond Model

Definitions

1. **Group survey:** Each group member is surveyed to discover their position on an issue, problem or topic. This process ensures that each member of the group is allowed to offer or state their point of view.
2. **Brain dump:** This is a great closer for students to sum up their knowledge on the topic(s) from the session. Have the students take out a blank piece of paper /Google Doc and write/type("dump") everything they know relevant from the discussion that day. The SI Leader can look at their work to assess the students' understanding of the course material and have them compare it to each other. It is important to at least discuss with the students so the SI Leader knows the students' comfort levels with the material.

Instructions (please provide aDETAILED activity breakdown below (and/or attached

1. Students are put into small groups and are asked to write down their confidence on a single notecard without names.
2. The cards are evidence of the group's confidence in the content.

Content (fully worked out problems & answers)

- No new worked problems; brain dump.

Checking for Understanding (questions & answers) continued...

- Review confidence levels
- Interpret the variety of confidence and their distribution in groups.

Plan Checklist

- ☐ Variety of Collaborative Learning Techniques and Learning Strategies
- ☐ Chapter/slide# where information is coming from
- ☐ Included timestamps (and buffer time as needed)
- ☐ Details on how leader will break up students
- ☐ Checking for Understanding questions
- ☐ Fully worked out problems and examples for each activity
- ☐ Necessary links required for online implementation (if applicable)
- ☐ Source material correctly cited (if applicable)
- ☐ Plan is uploaded to Teams